

Trained



Large labelled
training set

train



Corpus-specialized
features

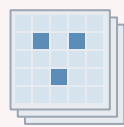
Standard



Cross-env.



Training-free, prior work

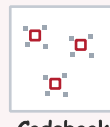


Frozen patch
features

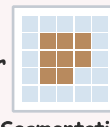
Standard



Cross-env.



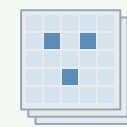
Codebook



Segmentation CLS attention

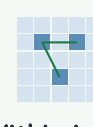
Prior-selected / aggregated features

Ours



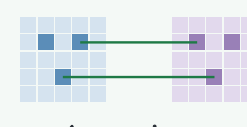
Frozen patch
features

Standard



Within-image

Balanced, matchable patch evidence



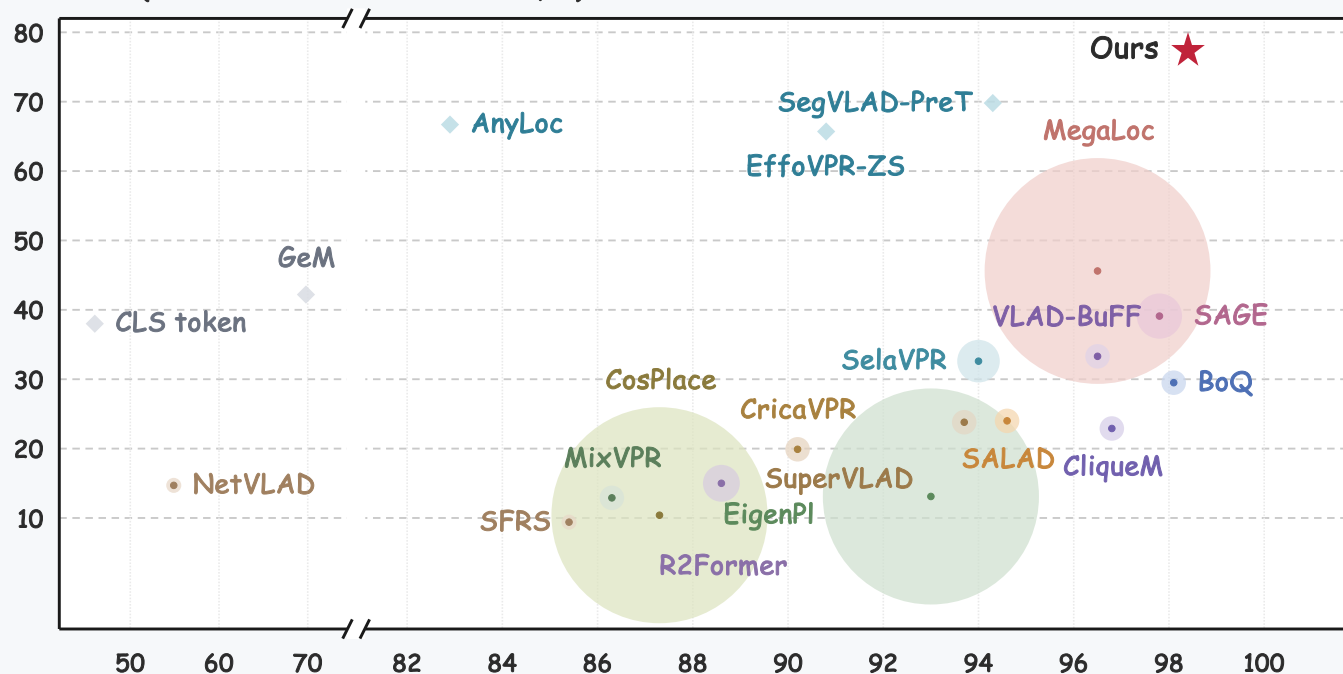
Across-image

Cross-env.



Training cost vs. performance

VP-Air R@1 (cross-environment benchmark, %)



★ ours (training-free)

♦ training-free

● trained

● 1 million images

● 5 million images

Performance across 17 benchmarks

